Python Lab: Hypothesis Testing

Statistical Computing Lab

Problem 1: One-Sample T-Test

Scenario: A restaurant industry standard claims that the average customer bill is exactly $18.00. Using the famous "tips" dataset, test whether the actual average total bill at this specific restaurant differs from the $18.00 standard at a 5% significance level.

Tasks

1. State the Null (H_0) and Alternative (H_a) hypotheses.
2. Perform a one-sample t-test comparing the bills to the $18.00 standard.
3. Print the t-statistic and p-value.
4. Conclude whether to reject the null hypothesis.

Problem 2: Independent Two-Sample T-Test (A/B Testing)

Scenario: The restaurant manager wants to know if there is a difference in tipping behavior between lunch and dinner services. Test if there is a statistically significant difference in the average tip amount between Lunch and Dinner.

Tasks

1. Check if the variances of the two groups are equal using Levene's test.
2. Perform the appropriate independent two-sample t-test (set equal_var based on step 1).
3. Interpret the p-value at $\alpha = 0.05$. Does service time affect average tip amounts?

Problem 3: Paired T-Test

Scenario: An exercise study tracked the heart rates (pulse) of individuals undergoing different activities. We want to test if there is a significant change in pulse from the 1-minute mark to the 15-minute mark for the same participants.

Tasks

1. Perform a paired t-test.
2. Calculate the average difference in pulse between the two timeframes.
3. State whether the heart rate significantly changed over the 15 minutes.

Problem 4: Chi-Square Goodness of Fit Test

Scenario: A historical maritime report claims that passenger class distribution on ocean liners of the era was roughly 25% First Class, 25% Second Class, and 50% Third Class. Using the Titanic dataset, check if the actual passenger class distribution on the Titanic matches this historical baseline.

Tasks

1. Calculate the array of expected_counts by multiplying the total number of passengers by the expected proportions.
2. Perform the Chi-Square Goodness of Fit test using chisquare.
3. Based on the p-value, write a conclusion stating whether the Titanic's class distribution fit the standard model of the era.